Restoring Ankle Push-off Dynamics in Markerless Gait Analysis by Calibrated Deconvolution

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Abstract

Markerless two-dimensional pose estimation reproduces hip and knee sagittal kinematics well but systematically under-reads the ankle range of motion (ROM), chiefly by flattening the fast plantar-flexion of push-off. We show that, on a joint-angle waveform, the pose estimator behaves as a subject-independent low-pass filter: it retains about 80% of the signal at 1 Hz but only ∼20% at 6–7 Hz. Modelling the estimator as a linear time-invariant system with transfer function $H(f)$, we estimate H once against synchronous optical motion capture (Vicon) on the BioCV dataset and invert it by Wiener deconvolution. The correction is applied as *mean restoration*: because deconvolution is linear, the systematic deficit lives in the mean cycle whereas the cycle-to-cycle spread is the clinical variability signal that must be preserved; we therefore deconvolve only the mean and add the resulting per-phase correction to every cycle. In a leave-one-subject-out validation over 9 subjects and 85 walking trials, the ankle ROM bias is halved ($-10.6° \to -5.3°$), the absolute ROM error drops from 10.8° to 6.7°, and inter-cycle variability is preserved exactly (1.73°). The method requires no motion capture at run time, makes no healthy-gait assumption, and improves every subject.

1 Introduction

Optical marker-based motion capture is the clinical reference for joint kinematics but is costly, lab-bound and slow to set up. Video-based *markerless* pipelines built on deep pose estimators—OpenPose [1], and more recently high-resolution whole-body models such as Sapiens [2]—promise the same measurement from a single camera [3]. Concurrent-validity studies now consistently report that hip and knee sagittal angles reproduce the optical reference to a few degrees, while the *ankle* is the weakest joint [4, 5]. The dominant error is a flattened push-off: the real foot plantar-flexes sharply to propel the body, but the tracked foot landmarks do not follow that fast motion (Figure 1).

Two families of remedies exist. Learned *marker augmenters* map sparse video keypoints to dense anatomical markers using a network trained on motion capture, as in OpenCap [6]; parametric body-and-foot mesh models regularise the pose with a learned shape prior. Both are powerful but opaque and data-hungry. Here we ask whether a simple, interpretable, *calibrated* signal-processing correction is enough. Our contributions are: (i) the observation that the estimator acts as a subject-independent low-pass filter on joint-angle waveforms; (ii) a cadence-adaptive Wiener deconvolution that inverts it from a filter calibrated once against Vicon; and (iii) a *mean-restoration* scheme that recovers the systematic amplitude while leaving inter-cycle variability—itself a clinical marker [7]—mathematically unchanged. We also report, for completeness, a set of geometric multi-view reconstructions that were tried and did not solve the ankle (Appendix A).

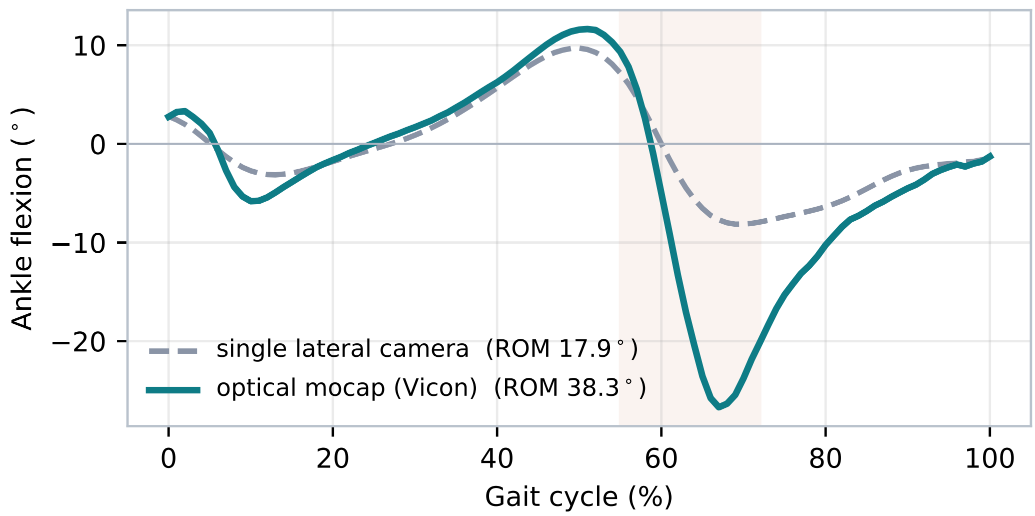

Figure 1: The deficit (subject P06). During push-off (shaded, ∼55–72% of the cycle) the optical reference reaches deep plantar-flexion, whereas the single lateral camera barely dips. The error is fast and phase-locked.

2 Method

2.1 Data and angle extraction

Video is processed by the `myogait` pipeline: per-axis normalisation, a 4 Hz Butterworth filter, 2-D sagittal joint angles, coordinate-based (Zeni) event detection, and time normalisation of each gait cycle to 101 points. The optical reference is the Visual3D-processed Vicon of the same trial, from which we recompute the ankle from the full 3-D markers. All angles are compared after removing the whole-cycle mean, so a definition offset between the two systems does not bias the comparison. Walking direction is resolved from several pan-invariant intra-frame body-orientation cues (foot-, nose- and toe-relative offsets) that remain valid when the camera pans to follow the subject, unlike the legacy hip-displacement heuristic. The final angle sign is then fixed by a flexion-positive canonicalisation that is itself independent of the detected direction; consequently the ankle metric reported here is invariant to the direction stage. We verified this directly: mirroring every landmark x-coordinate—the geometric equivalent of reversing the walking direction—reproduces the canonicalised ankle waveform to within 1°. Every number below is therefore identical whether the legacy or the newer detector is used.

2.2 The pose estimator as a linear filter

Write $c(\phi)$ for the true (Vicon) ankle angle over the normalised cycle ϕ and $v(\phi)$ for the markerless one. Taking the discrete Fourier transform over the cycle, the ratio between the video and reference spectra at each stride harmonic k is, across subjects, a smooth and reproducible function of frequency that decays with k (Figure 2). We therefore model the estimator as a linear time-invariant system,

$$V(f) = H(f)\,C(f), \tag{1}$$

where $H(f)$ is the complex transfer function of the tracker. Because H is a property of the estimator, not of the subject, it is estimated once and re-used. We sample it at the eight lowest harmonics with a Wiener (minimum mean-square) estimator over the training subjects,

$$\hat{H}_k = \frac{\langle V_k\,\overline{C_k}\rangle}{\langle |C_k|^2\rangle}. \tag{2}$$

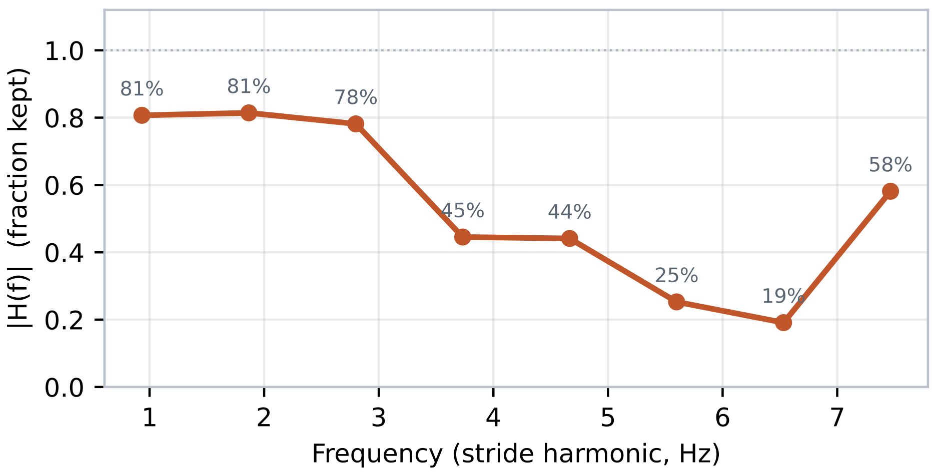

Figure 2: Estimated markerless transfer function $|H(f)|$: a low-pass. The tracker keeps ~80% of the ankle signal near 1 Hz but only ~20% by 6–7 Hz, exactly where the sharp push-off lives.

2.3 Cadence-adaptive inversion

At run time, for a trial of stride period T, harmonic k sits at frequency k/T; we interpolate H there, so the correction adapts to each subject's cadence and makes *no* assumption about a healthy gait pattern. The waveform is restored by regularised (Wiener) inverse filtering,

$$\hat{C}_k \;=\; V_k \,\frac{\overline{H_k}}{|H_k|^2+\lambda}, \qquad \lambda = 0.08, \tag{3}$$

the regulariser λ trading restoration against noise amplification.

2.4 Mean restoration

Deconvolution amplifies high-frequency noise; applied to each raw cycle it would inflate the very inter-cycle variability we wish to measure. Because the deconvolution operator $\mathcal{D}$ is linear, $\mathrm{mean}(\mathcal{D}\{c_i\}) = \mathcal{D}\{\bar{c}\}$: the systematic deficit lives in the *mean* cycle $\bar{c}$, while the deviations $c_i - \bar{c}$ carry the variability. We therefore deconvolve only the mean, form the per-phase correction, and add it to every cycle:

$$\Delta(\phi) = \mathcal{D}\{\bar{c}\}(\phi) - \bar{c}(\phi), \qquad c_i^{\mathrm{corr}}(\phi) = c_i(\phi) + \Delta(\phi). \tag{4}$$

Every cycle receives the *same* correction Δ, so the mean is restored to $\mathcal{D}\{\bar{c}\}$ while the cycle-to-cycle spread is left mathematically unchanged. This is the property that makes the method safe for variability-based clinical readings.

2.5 Geometric reconstructions considered

Before adopting deconvolution we tested whether the ankle deficit was *geometric*—recoverable with additional calibrated cameras—rather than a limitation of the landmarks themselves. Two ingredients were combined.

Skeleton locking (rigid-body constraints). Human segments are rigid. We first estimate each bone length as the temporal median $L_{ab} = \mathrm{median}_t \,\|X_a(t) - X_b(t)\|$ and rebuild the limb by chaining unit segment directions scaled by L_{ab} from a triangulated pelvis root, so no frame can stretch or collapse a bone. The foot is treated as one rigid triangle (ankle–heel–toe): at each frame we fit the canonical foot template $\{p_j\}$ to the observed points $\{q_j\}$ by a rigid transform (a Procrustes / Kabsch solution),

$$(R^\star, t^\star) = \arg\min_{R\in SO(3),\, t} \sum_j \left\| R p_j + t - q_j \right\|^2, \tag{5}$$

Table 1: Ankle metrics, leave-one-subject-out (9 subjects, 85 trials). Degrees.

Metric (ankle)	Before	**After**
|ROM error| vs Vicon	10.76	**6.71**
ROM bias (video − Vicon)	−10.61	−5.29
Waveform RMSE vs Vicon	3.70	**3.43**
Inter-cycle SD (preserved)	1.73	**1.73**
Inter-patient SD (Vicon 5.18)	3.63	4.75

and read the foot orientation off $R^{\star}$. This removes the non-rigid jitter of the foot landmarks (their inter-marker distance was seen to vary by 20–26% in video versus 2–5% in Vicon), which cleans the ankle waveform (shape correlation with Vicon rose from 0.84 to 0.88) but does *not* restore the missing amplitude.

Multi-view directional fusion. A lateral camera constrains a segment's antero-posterior and vertical components but is nearly blind to depth; a near-orthogonal (frontal) camera constrains the complementary axis. For a segment 3-D direction d, each camera c contributes one linear constraint: the observed 2-D direction u_c must be parallel to $M_c\, d$, with M_c the two in-plane rows of the camera rotation, i.e. $\left(u_c^{\perp\top} M_c\right) d = 0$. Two cameras give a 2×3 homogeneous system whose null space is d; chaining these directions with locked bone lengths yields a 3-D skeleton without triangulating the (noisy) foot position directly.

Both were evaluated but, as Appendix A shows, multi-view fusion *degraded* the ankle relative to a single lateral view—the frontal camera tracks the foot poorly—so the reported method uses one lateral camera plus the calibrated deconvolution.

3 Validation protocol

We use the BioCV dataset (University of Bath Research Data Archive [9]): synchronous Sapiens-2 markerless video (lateral camera) and Visual3D Vicon, 9 subjects, 85 walking trials. The transfer function is estimated **leave-one-subject-out**: for each test subject H is learned only on the others, so no test data enters the filter. Per trial and per side we report, for the ankle: the absolute ROM error and the signed bias versus Vicon; the waveform RMSE (mean-referenced); the inter-cycle SD (mean over phase of the standard deviation across a trial's cycles); and, across subjects, the inter-patient SD of the mean ROM. Agreement is summarised by Bland–Altman bias and $\pm 1.96\,$SD limits of agreement (LoA).

4 Results

Figure 3 overlays, for every one of the nine subjects, the raw and corrected mean cycle (with the corrected inter-cycle band) and the Vicon mean. The corrected curve deepens push-off toward the reference in every subject, and—crucially—the corrected band has the same width as the raw band. Table 1 gives the pooled metrics: the ROM bias is halved and the absolute error drops by 38%, while the inter-cycle SD is unchanged to two decimals. The Bland–Altman plots (Figure 4) show the bias moving toward zero and the lower limit of agreement tightening. The correction improves *every* subject (Figure 5), and the between-subject spread of ankle ROM moves *toward* the Vicon value ($3.6^\circ \to 4.7^\circ$, Vicon 5.2°): real inter-patient differences are recovered, not smoothed away.

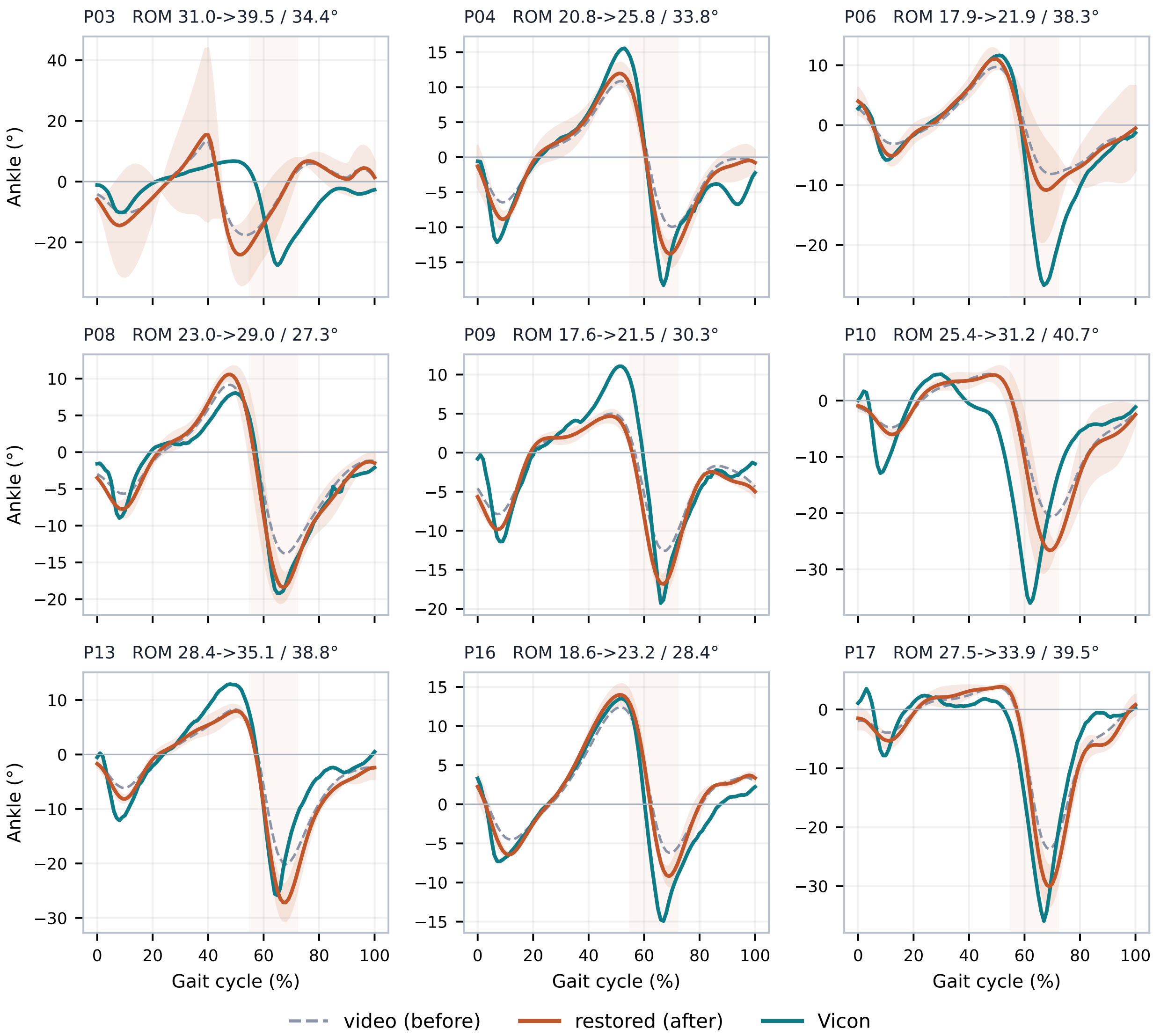

Figure 3: All nine subjects (per panel: raw ROM → restored / Vicon). Raw video (grey, dashed), restored mean ± inter-cycle SD (orange), Vicon (teal). The restored curve deepens push-off toward Vicon in every subject while the band width is preserved.

5 Discussion

The correction halves a systematic, phase-locked bias with a single lateral camera, one calibrated filter and about ten lines of code at run time. Three properties make it suitable for clinical, and specifically pathological, use. *Variability is preserved* by construction, so gait variability is neither smoothed nor inflated. *Inter-patient differences are restored, not erased*, which matters when the between-patient signal is itself the finding. And the filter is applied in Hz at the subject's own cadence, so there is *no healthy-gait prior* that could hallucinate a normal pattern onto a pathological one. It generalises: leave-one-subject-out, all nine subjects improved.

Limitations. (i) The residual bias (∼5°) is information the estimator never captured; a linear inverse filter can amplify what remains but cannot recreate what is absent—closing the gap further would require a learned non-linear augmenter [6]. (ii) The correction restores *amplitude* (ROM); the point-wise RMSE improves only modestly, as some shape error persists. (iii) H was estimated on one tracker and one dataset of healthy adults; a different tracker or population should re-estimate H (the calibration, not the method, is specific). (iv) Multi-view fusion did not help the ankle (Appendix A), so a single lateral view plus this correction is the recommended configuration. A definitive clinical

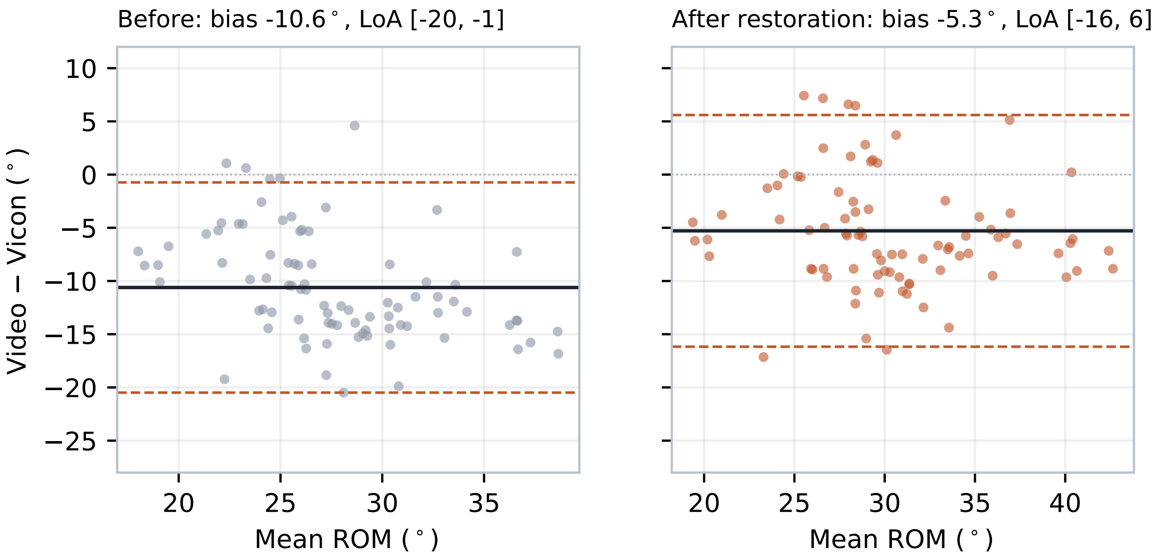

Figure 4: Bland–Altman on ankle ROM. The bias (solid) moves toward zero and the lower limit of agreement (dashed) tightens after restoration.

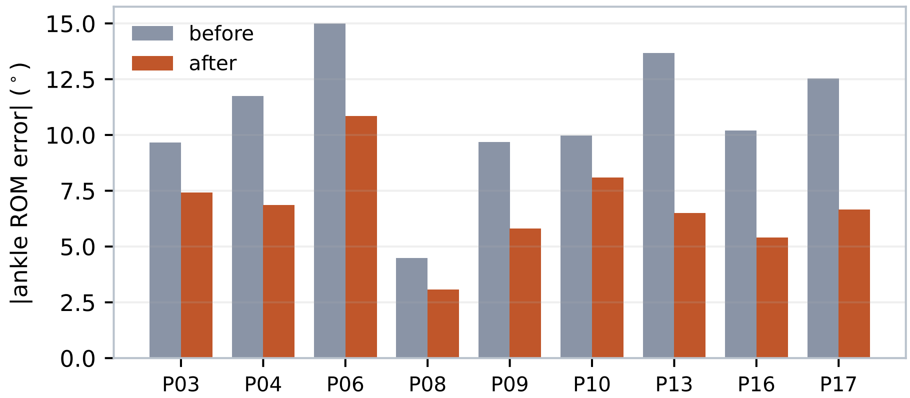

Figure 5: Per-subject absolute ankle ROM error, before and after. Every subject improves.

claim will require concurrent validation on a pathological cohort.

6 Conclusion

Treating a markerless pose estimator as a calibrated linear system and inverting it—while restoring only the systematic mean—recovers a clinically meaningful part of the ankle push-off that 2-D video loses, without motion capture at run time and without touching gait variability. The method ships in `myogait` as the opt-in `analyze_gait(restore_ankle_dynamics=True)`.

A Alternative reconstructions and ablation

On the one subject with four calibrated views plus Vicon, we scored ten multi-view reconstruction strategies by mean absolute ROM error against Vicon (Figure 6). Weighted or averaged fusion never beat the best single view; the strongest variant fused orthogonal image axes with locked bone lengths (Section 2.5), yet even it plateaued, because the push-off is largely absent from the foot landmarks—no geometric method fully recovers it.

The step-by-step ablation on the ankle (Table 2) shows that deconvolution on a single lateral camera is the best single step; multi-camera fusion degrades the ankle and the methods do not stack. The

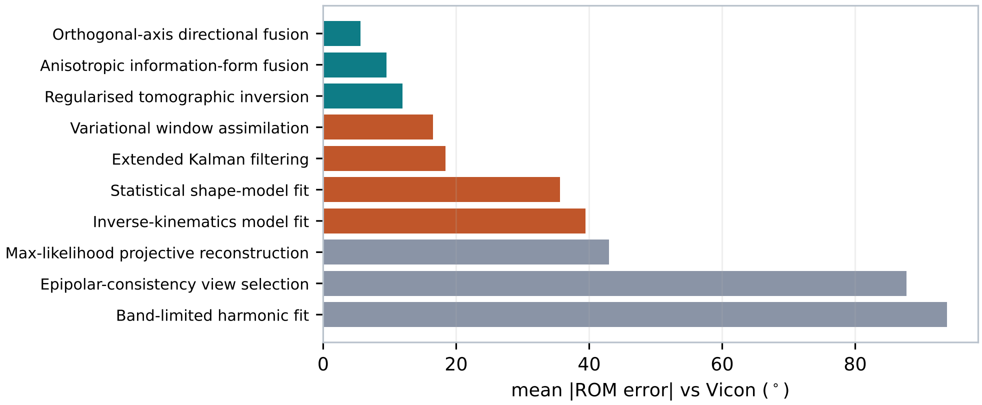

Figure 6: Ten multi-view reconstruction strategies (subject P06), mean |ROM error| vs Vicon. Teal: clear improvement (geometry-limited plateau); grey: failed.

raw per-cycle deconvolution used in this ablation inflates the inter-cycle SD; the mean-restoration variant of the main text removes that side-effect while keeping the accuracy gain.

Table 2: Ablation on the ankle (subject P06; Vicon ROM $\approx 38°$). Degrees.

Configuration	ROM	RMSE	inter-cycle SD
Single lateral camera	24.7	4.40	4.71
+ deconvolution (this work)	**32.7**	**3.37**	**5.84**
Multi-camera reconstruction	20.8	5.80	5.70
+ skeleton/foot locking	20.0	5.77	5.34
+ locking + deconvolution	23.9	6.35	6.55

Figure 7 shows the correction *effect* of each alternative on the mean ankle waveform: only the deconvolution (orange) deepens the push-off toward Vicon, whereas the multi-camera reconstruction and skeleton/foot locking leave it as flat as—or flatter than—the single camera. This is the visual counterpart of the table: geometry does not restore the missing amplitude, the calibrated filter does.

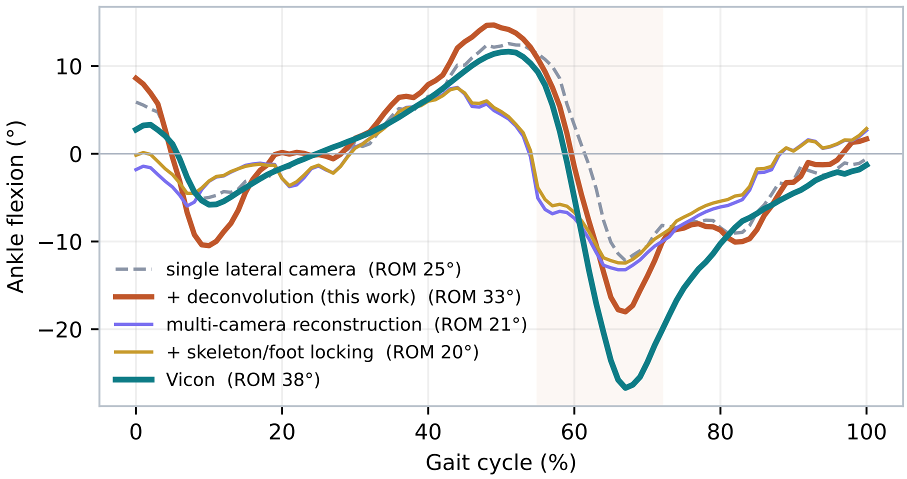

Figure 7: Effect of each alternative on the mean ankle waveform (subject P06, foot-flat referenced). Only the deconvolution (this work) restores the push-off toward Vicon; multi-camera reconstruction and skeleton/foot locking do not.

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